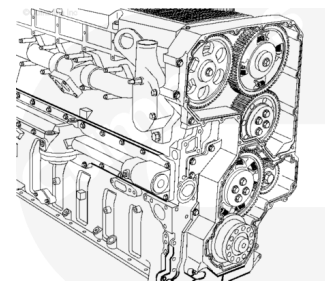


Trainings ▾

General Information

Front Gear Assembly Configuration

Two front gear train combinations are presently utilized on the Signature™, ISX, and QSX15 engines. Signature™, ISX3, ISX2, and QSX15 engines use a scissor gear for the lower idler gear and the injector cam gear. ISX1 does **not** use a scissor gear at the idler location.



00c00035

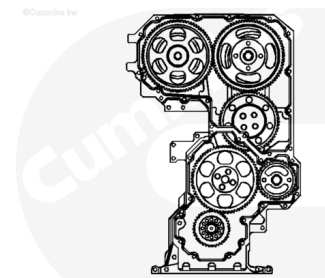
ISX2 engines built between January 1999, and January 2000, utilize a scissor gear for the lower idler gear **only**. The injector cam gear is a straight cut spur gear, similar to the ISX1. After January, 2000, the ISX2 has scissor gears for both the cam and lower idler.

This procedure describes the removal, inspection, and installation of the different gear combinations.

Front Gear Assembly without Scissor Gear

View shown is from left to right and top to bottom.

- Valve camshaft gear
- Injector camshaft gear
- Adjustable idler gear
- Lower idler gear
- Accessory gear
- Crankshaft gear.

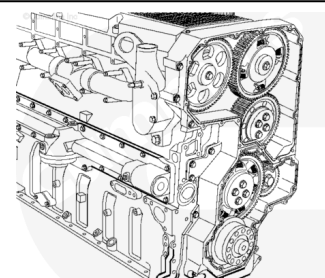


01c00349

Front Gear Assembly with Scissor Gear

View shown is from left to right and top to bottom.

- Valve camshaft gear
- Injector camshaft scissor gear
- Adjustable idler gear
- Lower idler scissor gear
- Accessory gear
- Crankshaft gear.

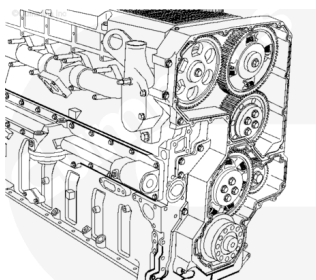


00c00035

Scissor Gear Definitions

⚠ WARNING ⚠

Do not attempt to remove any gears before reading scissor gear definitions. Serious personal injury or engine damage can result if instructions are not followed.



00c00035

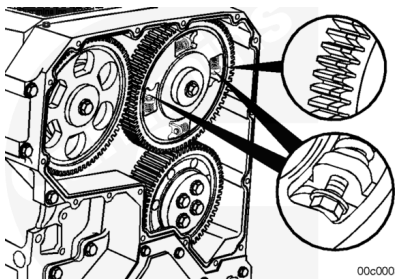
The following terms describe the conditions of the scissor gears for removal, installation, and operation.

Unloaded

- The gear will be unloaded when removing, installing, and setting gear backlash.

Unload the gear by backing out two gear adjusting screws until the gear teeth align.

The idler scissor gear is loaded when the gear backlash is set.

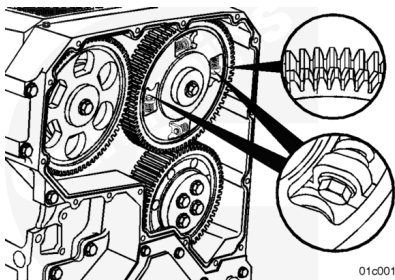


00c00037

Loaded

- The gear teeth are loaded (functional) and are spread between the mating gear teeth.

The gear is loaded by tightening all the adjusting screws. The gear teeth will be spread and no movement will be felt against the valve cam gear. The gear will be in the loaded condition for engine operation.



01c00111

Initial Check

Note : Use the following steps to check base engine timing.

Timing Wedges:

The timing code is located on the engine dataplate.

- First number - injector wedge.
- Second number - valve wedge.

©Cummins Inc.

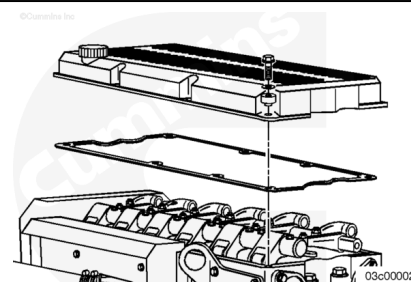
Engine Cert Id.	Displacement	15L	CPL
	Fuel Rate at Rated Power		
Timing T.D.C. (inj/valve)	4/6		Idle Speed (RPM)
Valve Lash Cold (in/mm)	Int.		

1 2

07c00169

If the engine dataplate is missing, a new dataplate should be ordered. Use the following procedure to identify the timing code for an engine with a missing dataplate. Refer to Procedure 850-029 in Section V. ([/qs3/pubsys2/xml/en/procedures/99/99-850-029.html](http://qs3/pubsys2/xml/en/procedures/99/99-850-029.html))

Remove the rocker lever cover. Refer to Procedure 003-011 in Section 3. ([/qs3/pubsys2/xml/en/procedures/10/10-003-011-tr.html](http://qs3/pubsys2/xml/en/procedures/10/10-003-011-tr.html))

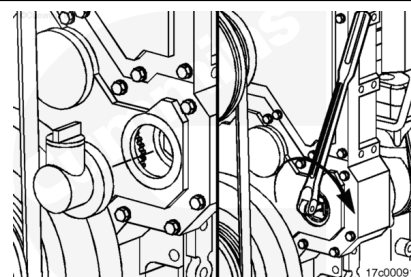


Note : Use this step if the engine is equipped with an air compressor:

Remove the oil fill connector from the lower gear case cover.

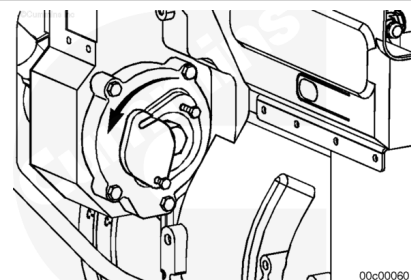
Insert a 3/4-inch drive ratchet and extension into the air compressor drive.

Rotate the air compressor drive **clockwise**, as viewed from the front of the engine.

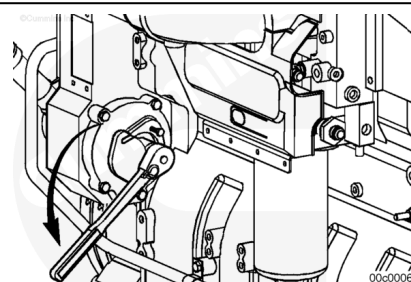


Note : Use this step if the engine is not equipped with an air compressor:

Loosen the cover plate capscrews and rotate the cover or remove the oil fill tube, if equipped.

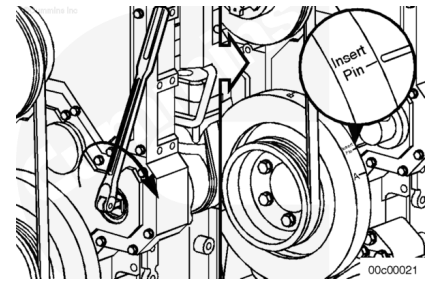


Use a 1½ inch socket to push the barring gear into the gear mesh. Rotate the barring adapter to bar the engine.



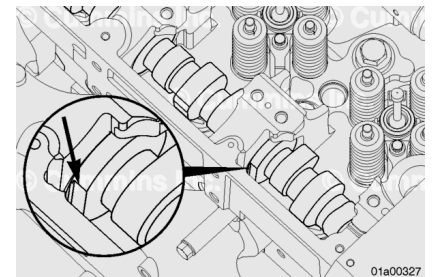
Rotate the crankshaft **clockwise** until the insert pin mark on the outside diameter of the engine vibration damper is aligned with the mark on the gear housing.

Q SX15 engines are equipped with dual vibration dampers. The crank pin mark is located on the damper closest to the engine.



Verify that the wedge flats of the camshaft are visible when viewed from above.

If wedge flats are **not** visible when at insert pin, the engine is **not** on the timing stroke (timing is performed at cylinder number 1 compression stroke). Rotate the crankshaft 360 degrees. This will change the orientation of the camshaft 180 degrees and the wedge flats will be visible from above. This indicates cylinder number 1 is on the compression stroke.



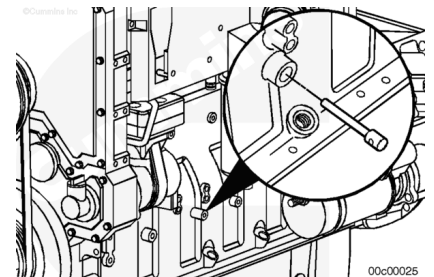
Remove the plug from the timing pin boss.

Lock the crankshaft in place with the crankshaft locking pin, Part Number 3163020.

The pin **must** be in good condition; **not** bent, mushroomed, oval, etc.

When properly installed, the colored band on the timing pin will be lined up with the surface of the timing pin boss of the block.

The pin is **not** correctly seated in the crank notch if the green band is either completely visible outside the block, or is **not** visible at all.



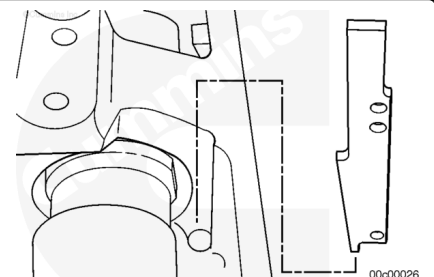
⚠ CAUTION ⚠

Do not drive the wedges into place. Component damage can occur. Install wedges by hand.

Insert the timing wedges listed on the dataplate. The wedges **must** be in good condition, with smooth flats and spine. Using damaged wedges can result in poor timing. Replace worn and damaged wedges.

Proper fits should leave no gap between the wedge face and cam, or between the back of the wedge and the head.

Not all wedges have lines; if the wedge has a line, the line should be level with the top of the head/cover rail.



For properly fitted wedges with lines, timing variation represented by the position of the line is small.

Note : If the line is high for CM871 engines:

Valve Cam: No performance complaints arise from valve timing variation of this amount. The line on a valve cam wedge should be disregarded.

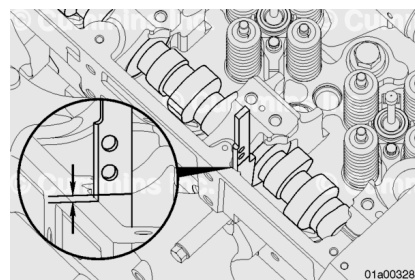
Injector Cam: Timing may need to be physically measured to address potential performance issues if they exist. Review engine history for performance complaints including filter plugging, white smoke, etc.

Discuss the next steps with appropriate Cummins Inc. technical support resource if engine has history of these complaints and the line on the injector wedge is higher than the head cover rail.

Note : If the line is high for all other 15L engines (except CM871):

Both the valve and injection timing is robust to timing variation of this amount, the line should be disregarded.

Use the following procedure if static injection timing requires further measurement or adjustment. Refer to Procedure 006-025 in Section 6. (</qs3/pubsys2/xml/en/procedures/10/10-006-025.html>)



Preparatory Steps

⚠ WARNING ⚠

Batteries can emit explosive gases. To reduce the possibility of personal injury, always ventilate the compartment before servicing the batteries. To reduce the possibility of arcing, remove the negative (-) battery cable first and attach the negative (-) battery cable last.

- Disconnect the batteries. See equipment manufacturer service information.

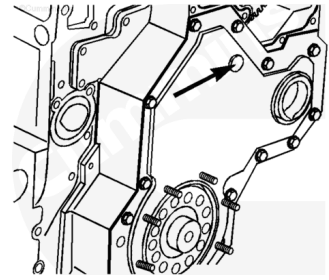
⚠ CAUTION ⚠

The timing pin and wedges must be installed before any gears are removed and must remain in place until all gears are fully installed and tightened. Failure to install the timing pin and wedge can lead to improper timing of the engine, resulting in severe engine damage or engine failure.

Note : Perform Initial Check section before removing gears.

Removal of the lower cover is **not** required when setting the gear backlash on engines that utilize all non-scissor gears. An access hole is provided to insert a shim between the lower non-scissor idler gear and the adjustable idler gear. The access hole is sealed with a plug.

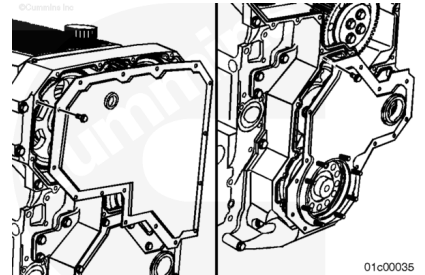
It will be helpful to push the gear cover plug out from the rear. It is **not** necessary to remove the lower gear cover to push out the plug.



01c00337

Remove the upper gear cover. Refer to Procedure 001-079 in Section 1. (</qs3/pubsys2/xml/en/procedures/10/10-001-079-tr.html>)

Remove the lower gear cover, if needed. Refer to Procedure 001-080 in Section 1. (</qs3/pubsys2/xml/en/procedures/10/10-001-080-tr.html>)



01c00035

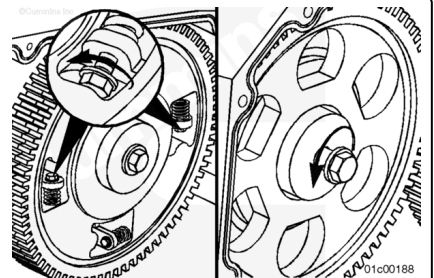
Remove

Valve Camshaft Gear

The injector camshaft scissor gear **must** be unloaded during removal of the valve camshaft gear. To unload the gear, two of the gear screws **must** be backed out far enough to allow the teeth on the scissor gear to be aligned.

Unload the injector camshaft scissor gear.

Remove the valve camshaft gear mounting capscrew and washer.

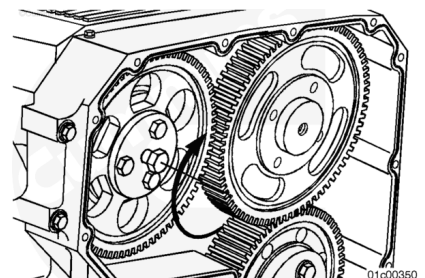


01c00188

Early production engines were **not** equipped with puller holes in the valve camshaft gear. A three-jaw gear puller **must** be used for gears that do **not** have puller holes. The injector and valve gear puller, Part Number 3163069, can be used for gears that have puller holes.

Install the gear puller, Part Number 3163069.

Install the valve camshaft gear mounting capscrew completely.



01c00350

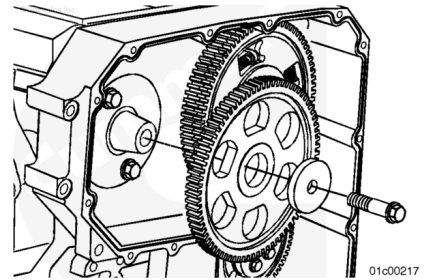
Loosen the capscrew two turns.

Use the gear puller to remove the valve camshaft gear from the camshaft nose.

Remove the puller.

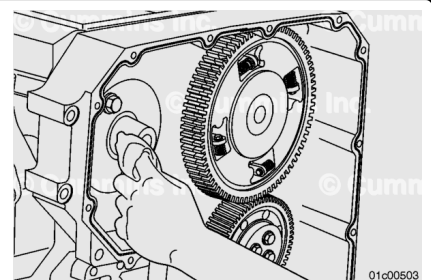
Do **not** completely remove the camshaft gear mounting capscrew when the three-jaw puller is used. The capscrew serves as a safety catch by preventing the gear from shooting off the camshaft nose.

Remove the valve camshaft gear mounting capscrew, spacer, and valve camshaft gear.



⚠ WARNING ⚠

When using solvents, acids, or alkaline materials for cleaning, follow the manufacturer's recommendations for use. Wear goggles and protective clothing to reduce the possibility of personal injury.



⚠ WARNING ⚠

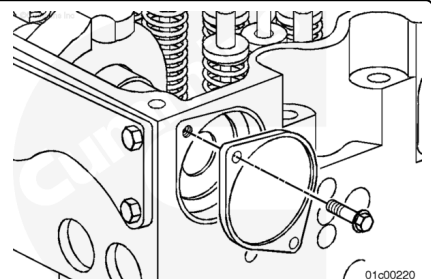
Some solvents are flammable and toxic. Read the manufacturer's instructions before using.

Use a lint-free cloth and solvent to clean the Loctite™ residue from the nose of the camshaft.

Injector Camshaft Scissor Gear

⚠ CAUTION ⚠

Remove the rear injector camshaft cover plate before removal of the injector camshaft gear or camshaft. The force required to remove the camshaft gear creates sufficient backward movement of the camshaft to crack the rear cover plate.

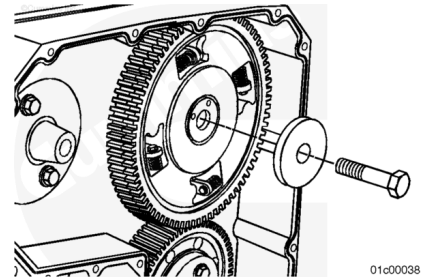


Remove the injector camshaft rear cover plate.

Remove the injector camshaft scissor gear mounting capscrew and washer.

Install gear puller, Part Number 3163069.

Install the injector camshaft scissor gear mounting capscrew.
Loosen the capscrew one turn.

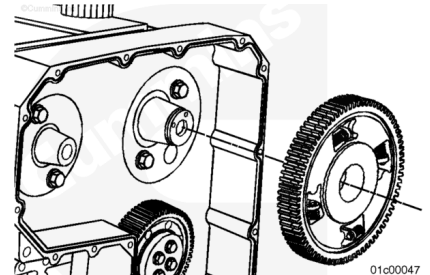


Use the gear puller to force the injector camshaft scissor gear off the camshaft taper.

Remove the gear puller.

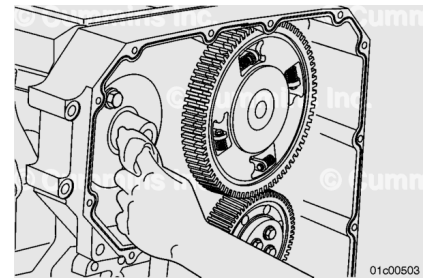
Remove the capscrew.

Remove the injector camshaft scissor gear.



⚠ WARNING ⚠

When using solvents, acids, or alkaline materials for cleaning, follow the manufacturer's recommendations for use. Wear goggles and protective clothing to reduce the possibility of personal injury.



⚠ WARNING ⚠

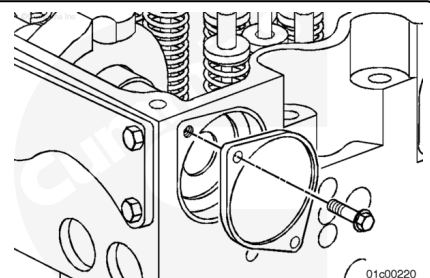
Some solvents are flammable and toxic. Read the manufacturer's instructions before using.

Use a lint-free cloth and solvent to clean the Loctite™ residue from the nose of the camshaft.

Injector Camshaft Non-Scissor Gear

⚠ CAUTION ⚠

Remove the rear injector camshaft cover plate before removal of the injector camshaft gear or camshaft. The force required to remove the camshaft gear creates sufficient backward movement of the camshaft to crack the rear cover plate.

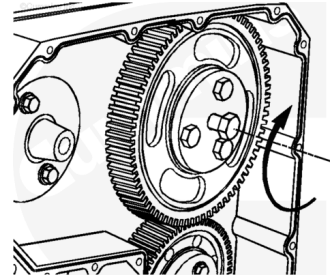


Remove the injector camshaft rear cover plate.

Remove the injector camshaft gear mounting capscrew and washer.

Install gear puller, Part Number 3163069.

Install the injector camshaft gear mounting capscrew. Loosen the capscrew one turn.



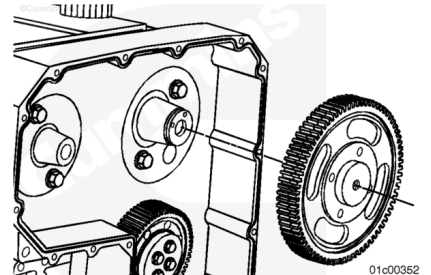
01c00351

Use the gear puller to force the injector camshaft gear off the camshaft taper.

Remove the gear puller.

Remove the capscrew.

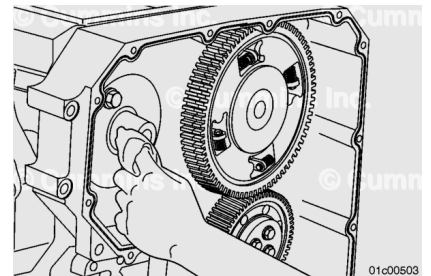
Remove the injector camshaft gear.



01c00352

⚠ WARNING ⚠

When using solvents, acids, or alkaline materials for cleaning, follow the manufacturer's recommendations for use. Wear goggles and protective clothing to reduce the possibility of personal injury.



01c00503

⚠ WARNING ⚠

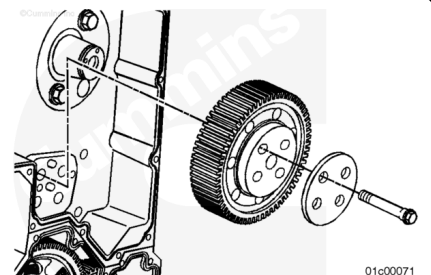
Some solvents are flammable and toxic. Read the manufacturer's instructions before using.

Use a lint-free cloth and solvent to clean the Loctite™ residue from the nose of the camshaft.

Adjustable Idler Gear

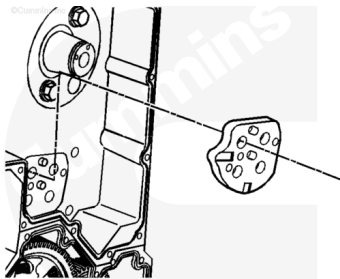
The adjustable idler gear shaft can fall out when the adjustable idler gear is removed.

Remove the retainer capscrews, retainer, gear, and adjustable idler gear shaft.



01c00071

Remove the mounting spacer.



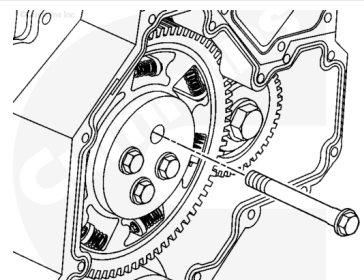
01c00048

Lower Idler Scissor Gear

The lower idler scissor gear **must** be unloaded during removal. To unload the gear, two of the gear screws **must** be backed out far enough to allow the teeth on the scissor gear to be aligned.

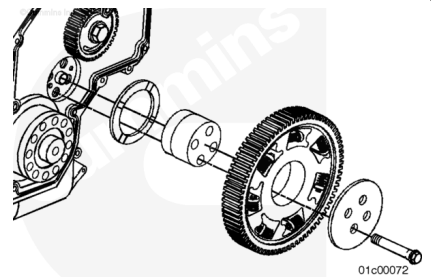
Unload the lower idler scissor gear.

Remove the idler scissor gear capscrews.



01c00045

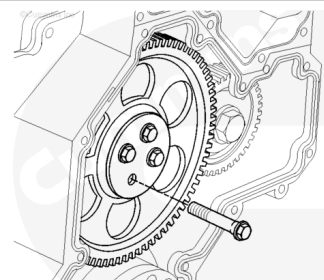
Remove the retainer, gear, shaft, and thrust bearing.



01c00072

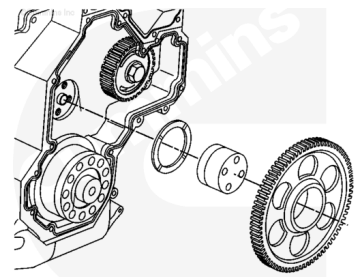
Lower Idler Non-Scissor Gear

Remove the idler non-scissor gear capscrews and retainer.



01c00353

Remove the gear, shaft, and thrust bearing.



01c00354

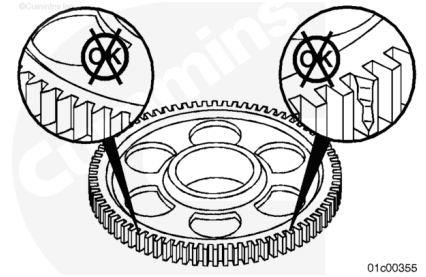
Inspect for Reuse

Inspect the inner diameter bushings for heavy wear or pitting. For bushings with a clinch joint on the inner diameter, inspect the connection for any separation or delamination that exposes underlying steel.

Inspect the gear teeth on all gears for pitting, heavy wear, or breakage.

Inspect the gear for cracking around the puller holes.

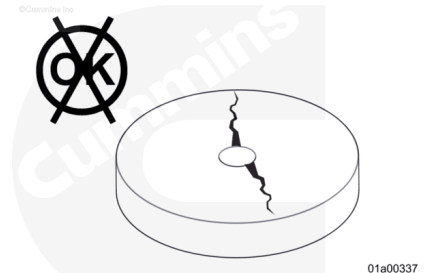
Replace any pitted, heavily worn, or damaged gear, and inspect the mating gear for associated damage.



Note : There are two types of gear retaining washer, powdered metal and machined-from-solid. Powdered metal retaining washers have a profiled surface and are **not** to be reused. Machined-from-solid washers have a flat surface and can be reused, if no damage is found during inspection.

Inspect the machined-from-solid gear retaining washer for cracking or other damage.

Replace the gear retaining washer if any damage is found.

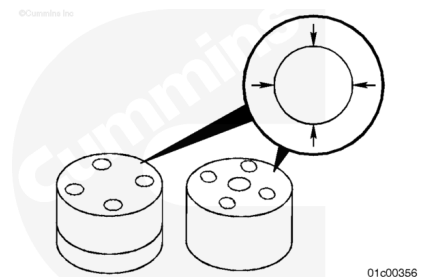


Inspect the lower idler shaft, and adjustable idler shaft for heavy wear.

Measure the idler shafts' outside diameter.

Lower Idler Shaft Outside Diameter		
mm		in
88.88	MIN	3.4992
88.89	MAX	3.4997

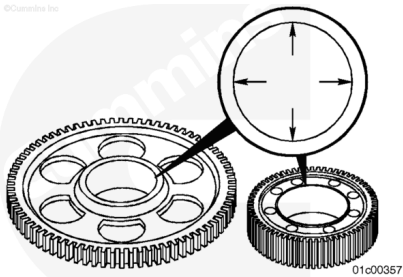
Adjustable Idler Shaft Outside Diameter		
mm		in
88.88	MIN	3.4992
88.89	MAX	3.4997



Inspect the lower idler gear bore and adjustable idler gear bore for heavy wear.

Measure the inside diameter of the lower idler and adjustable idler gear bore.

Note : : If the gear bushing replacement is necessary, purchasing the assembled gear and bushing is recommended. Gear bushings are available separately but require final machining of the inside diameter once installed into the gear.



Lower Idler Gear Bore Inside Diameter

mm		in
88.962	MIN	3.5024
88.988	MAX	3.5035

Adjustable Idler Gear Bore Inside Diameter

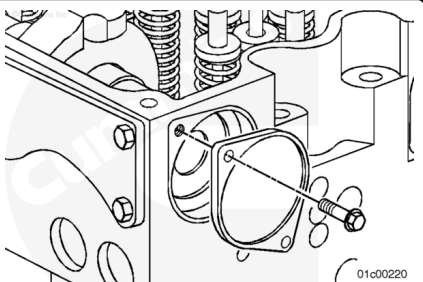
mm		in
88.962	MIN	3.5024
88.988	MAX	3.5035

Inspect the thrust bearing(s) for pitting, cracking, scratching, uneven wear, or other damage. Replace the thrust bearing(s) and inspect the corresponding gear, if damage is found.

Install

If removed, install the injector camshaft and valve camshaft rear cover plates.

Torque Value: 40 n•m [30 ft-lb]



Verify the correct wedges are installed and properly fitted as described in the Initial Check section.

Engine Cert Id.	Displacement	15L	CPL
	Fuel Rate at Rated Power		
Timing T.D.C. (inj/valve)	4/6	Idle Speed (RPM)	
Valve Lash Cold (in/mm)		Int.	

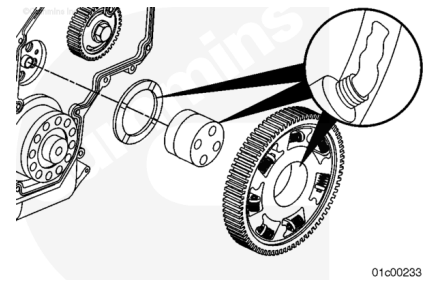
1 2

Lower Idler Scissor Gear

The lower scissor idler gear **must** be unloaded during installation. To unload the gear, two of the gear screws **must** be backed out far enough to allow the teeth on the scissor gear to be aligned.

Apply Lubriplate™ to the thrust bearing, shaft, and lower scissor idler gear.

Install the thrust bearing, shaft, and lower scissor idler gear. The shaft **must** be installed with the word “TOP” (stamped on the end of the shaft) toward the top of the engine.



01c00233

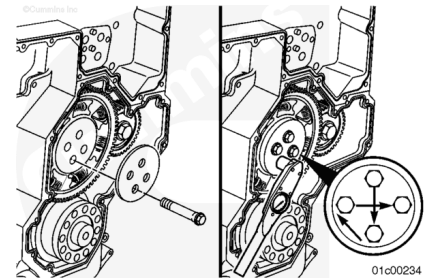
The slots in the gear retainer plate go toward the scissor gear.

Install the retainer and capscrews.

Tighten all capscrews in a star pattern.

Torque Value:

1. 30 n•m [22 ft-lb]
2. Rotate each capscrew an additional 60 degrees $\pm$ 5 degrees in the same alternating patterns.



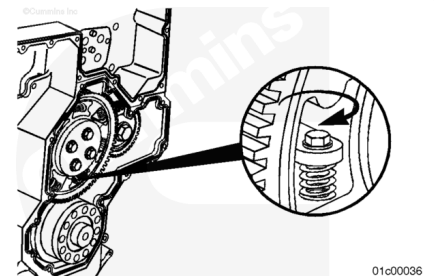
01c00234

⚠ CAUTION ⚠

Make sure all the scissor gear screws are tightened to the proper torque so that they do not back out during engine operation. Severe engine damage can occur.

Load the lower idler scissor gear by tightening all the gear screws.

Torque Value: 28 n•m [248 in-lb]

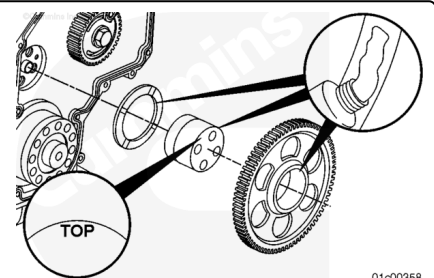


01c00036

Lower Idler Non-Scissor Gear

Apply Lubriplate™ to the thrust bearing, shaft, and lower non-scissor idler gear.

Install the thrust bearing, shaft, and lower non-scissor idler gear. The shaft **must** be installed with the word “TOP” (stamped on the end of the shaft) toward the top of the engine.



01c00358

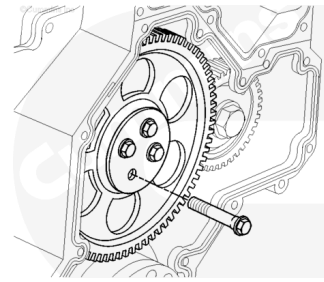
The slots in the gear retainer plate go toward the scissor gear.

Install the retainer and capscrews.

Tighten all capscrews in a star pattern.

Torque Value:

1. 30 N•m [22 ft-lb]
2. Rotate each capscrew an additional 60 degrees $\pm$ 5 degrees in the same alternating patterns.

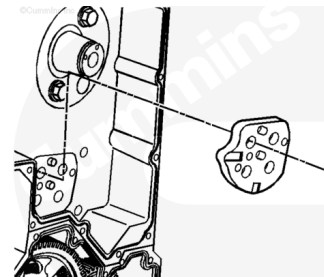


01c00353

Adjustable Idler Gear

The spacer plate is doweled to the block with two standard dowels.

Install the mounting spacer plate.

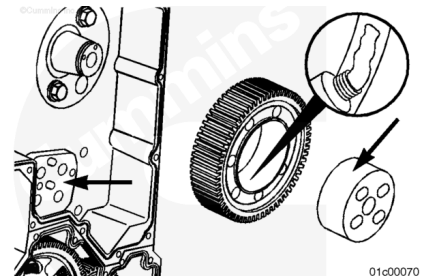


01c00048

Apply Lubriplate™ to the spacer, shaft, and adjustable idler gear.

It is possible to install the shaft in an incorrect orientation; however, it will **not** be possible to set the gear backlash correctly.

Install the spacer, shaft, and adjustable idler gear. If the upper idler gear idler shaft has the word "TOP" stamped on the front face, the upper idler gear idler shaft **must** be installed with the word "TOP" oriented toward the top of the engine. If the upper idler gear idler shaft has an arrow stamped on the face, the upper idler gear idler shaft **must** be installed with the arrow oriented in the 9 o'clock position.

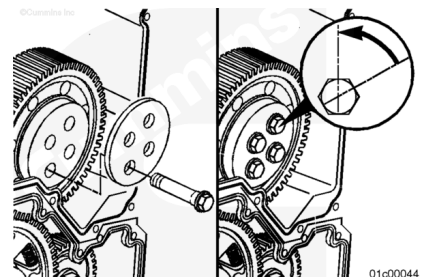


01c00070

The slots on the gear retainer plate go toward the gear.

The adjustable idler gear is installed loosely, because it is moved into its final position later in the timing process.

Install the retainer and capscrews. Hand-tighten the capscrews and then loosen by one "flat," which is 60 degrees.

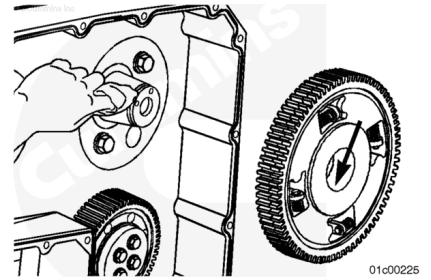


01c00044

Injector Camshaft Scissor Gear

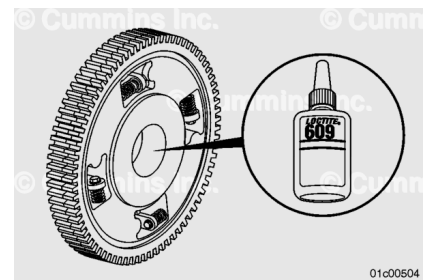
⚠ CAUTION ⚠

The injector camshaft nose and injector camshaft scissor gear must be clean and dry prior to assembly. Wipe off the taper surface with a clean, lint-free cloth and apply a thin coat of Loctite™ 609, Part Number 3823718, to the inside diameter of the camshaft gear. Use care not to contaminate the joint with oil prior to assembly or engine damage may occur.



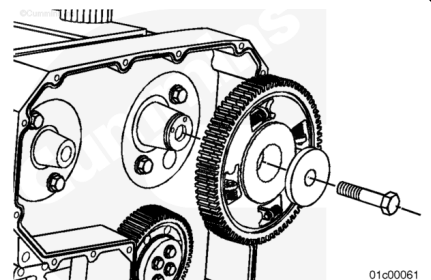
The injector camshaft scissor gear **must** be unloaded during installation. To unload the gear, two of the gear screws **must** be backed out far enough to allow the teeth on the scissor gear to be aligned.

Apply a thin coat of Loctite™ 609, Part Number 3823718, to the inside diameter of the camshaft gear.



⚠ CAUTION ⚠

There are two types of gear retaining washer, powdered metal and machined-from-solid. Powdered metal retaining washers have a profiled surface and are not to be reused. Machined-from-solid washers have a flat surface and can be reused if no damage is found during inspection.



The injector camshaft scissor gear needs to be fully engaged on the camshaft nose taper, yet loose enough so it can rotate on the camshaft nose.

Install the injector camshaft scissor gear loosely on the camshaft nose taper.

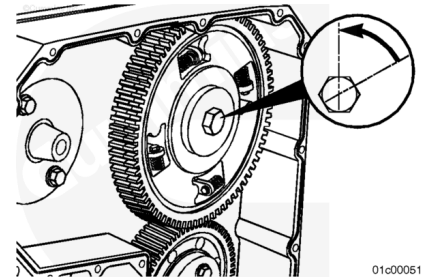
The injector camshaft scissor gear mounting washer is larger in diameter than the valve camshaft mounting washer.

Install the mounting washer and capscrew.

Hand-tighten the capscrew and then loosen by one "flat," which is 60 degrees.

The injector camshaft scissor gear is still loose at this point, to allow the adjustable idler gear to be properly positioned.

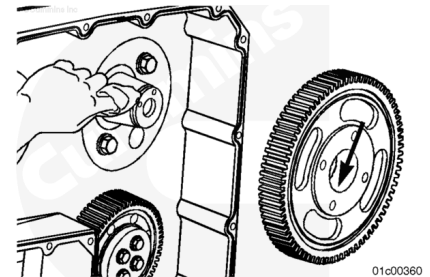
Do **not** seat the gear.



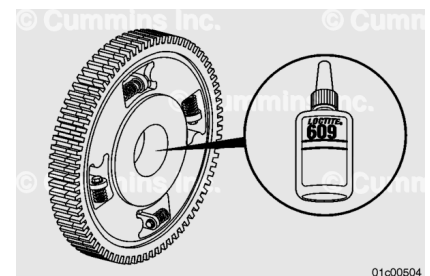
Injector Camshaft Non-Scissor Gear

⚠ CAUTION ⚠

The injector camshaft nose and injector camshaft non-scissor gear must be clean and dry prior to assembly. Wipe off the taper surface with a clean lint-free cloth and apply a thin coat of Loctite™ 609, Part Number 3823718, to the inside diameter of the camshaft gear. Use care not to contaminate the joint with oil prior to assembly or engine damage may occur.

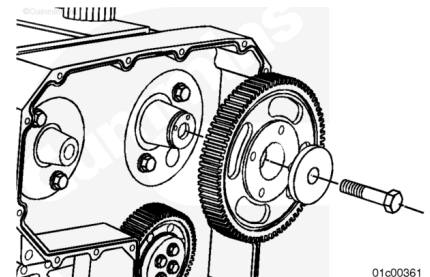


Apply a thin coat of Loctite™ 609, Part Number 3823718, to the inside diameter of the camshaft gear.



⚠ CAUTION ⚠

Powdered metal injector gear retaining washers, identified by profiled surface, are not to be reused, and must be replaced with a machined-from-solid retaining washer, identified by a flat surface. Machined-from-solid retainers can be reused if no damage is found during inspection.



The injector camshaft non-scissor gear needs to be fully engaged on the camshaft nose taper, yet loose enough so it can rotate on the camshaft nose.

Install the injector camshaft non-scissor gear loosely on camshaft nose taper.

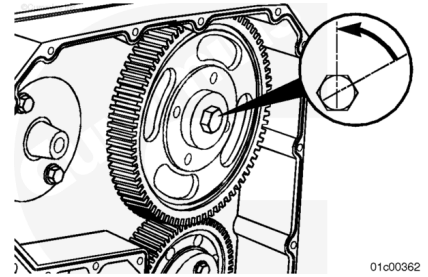
The injector camshaft non-scissor gear mounting washer is larger in diameter than the valve camshaft mounting washer.

Install the mounting washer and capscrew.

Hand-tighten the capscrew and then loosen by one "flat," which is 60 degrees.

The injector camshaft non-scissor gear is still loose at this point to allow the adjustable idler gear to be properly positioned.

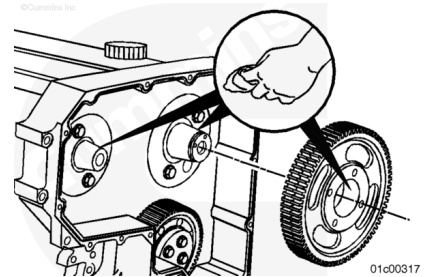
Do **not** seat the gear.



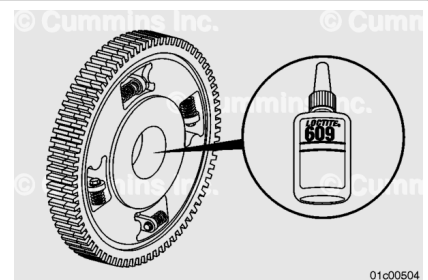
Valve Camshaft Gear

⚠ CAUTION ⚠

The valve camshaft nose and valve camshaft gear must be clean and dry prior to assembly. Wipe off the taper surface with a clean, lint-free cloth and apply a thin coat of Loctite™ 609, Part Number 3823718, to the inside diameter of the camshaft gear. Use care not to contaminate the joint with oil prior to assembly or engine damage may occur.



Apply a thin coat of Loctite™ 609, Part Number 3823718, to the inside diameter of the camshaft gear.

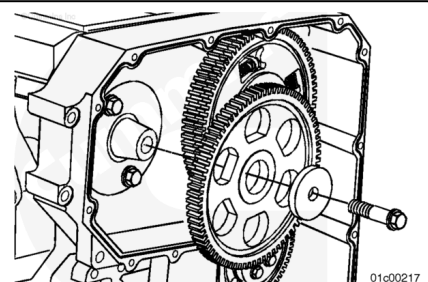


The valve camshaft gear needs to be fully engaged on the camshaft nose taper, yet loose enough so it can rotate on the camshaft nose.

Install the valve camshaft gear loosely on the camshaft nose taper.

The valve camshaft mounting washer is smaller in diameter than the injector camshaft scissor gear mounting washer.

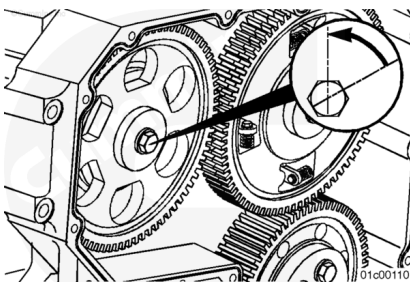
Install the mounting washer and capscrew.



Hand-tighten the capscrew and then loosen by one "flat," which is 60 degrees.

The valve camshaft gear is still loose at this point to allow the adjustable idler gear to be properly positioned.

Do **not** seat the gear.



Adjust

Backlash (Scissor Gear Combinations)

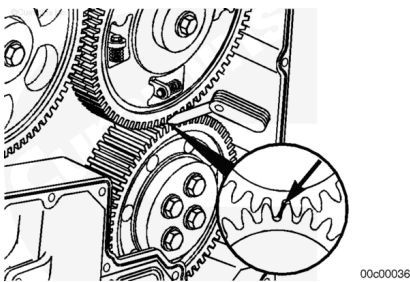
⚠ CAUTION ⚠

Do not set backlash between the adjustable idler gear and the lower scissor idler gear. Severe engine damage can occur.

This backlash step works with the scissor type injector camshaft gear and lower idler gear, and the non-scissor injector camshaft gear with a scissor type lower idler gear.

The injector camshaft scissor gear **must** still be unloaded during this step. To unload the gear, two of the gear screws **must** be backed out far enough to allow the teeth on the scissor gear to be aligned.

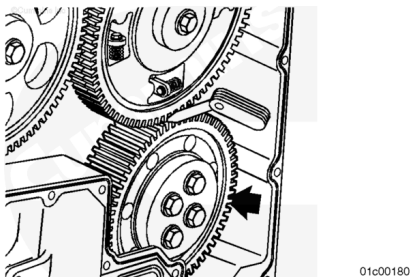
Insert, as shown, a narrow shim (0.25 mm [0.010 in]) into the gear teeth mesh between the adjustable idler gear and the **wide** (rear side) gear of the injector camshaft scissor gear.



Shim Tolerance

mm		in
0.225	MIN	0.009
0.275	MAX	0.011

Note : The gear backlash set procedure for engines that have a non-scissor injector camshaft gear and a scissor type lower idler gear is the same as that of engines that utilize two scissor type gears.



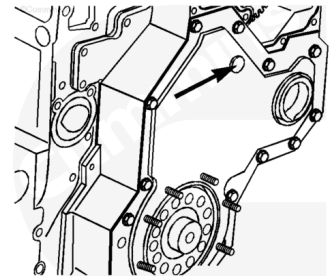
Use hand pressure, **only**, to move the adjustable idler gear toward the center of the engine (into mesh), so there is no gap on either side of the shim.

Both of the camshaft gears will rotate slightly on the camshafts during this step.

Slight drag will be felt on the shim while pulling the shim in and out.

Backlash (Non-Scissor Gear Combinations)

Removal of the lower front cover is **not** required when setting the gear backlash on engines that utilize all non-scissor gears. An access hole is provided to insert a shim between the lower non-scissor idler gear and the adjustable idler gear. The access hole is sealed with a plug. The plug can be pushed out from the rear side without removing the gear cover.



01c00337

The following procedure will be shown with the lower gear cover removed.

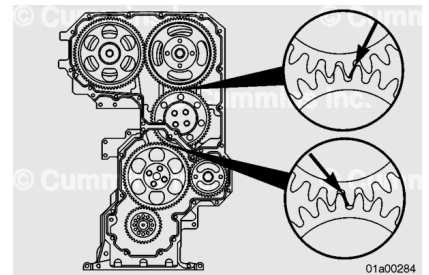
Insert, as shown, a shim 0.25 mm [0.010 in] into the gear teeth mesh between the adjustable idler gear and the injector camshaft gear.

Insert, as shown, a shim 0.25 mm [0.010 in] into the gear teeth mesh between the adjustable idler and the lower non-scissor idler gear.

Use hand pressure, **only**, to move the adjustable idler gear so there is no gap between the teeth and either side of the shims.

Both of the camshaft gears will rotate slightly on the camshafts during this step.

Slight drag will be felt on the shim while pulling the shim in and out.



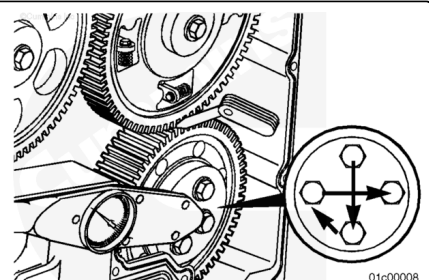
01a00284

Adjustable Idler Gear

While holding the adjustable idler gear in position, tighten the adjustable idler gear mounting capscrews.

Torque Value:

1. 88 n•m [65 ft-lb]
2. Rotate each capscrew an additional 60 degrees in the same sequence as above.



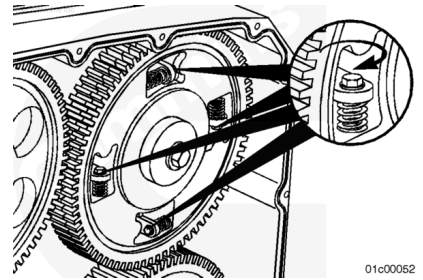
01c00008

Remove the feeler gauge(s).

Injector Camshaft Scissor Gear

⚠ CAUTION ⚠

Make sure all the scissor gear screws are tightened to the proper torque so that they do not back out during engine operation. Severe engine damage can occur.



Load the injector camshaft scissor gear by tightening all the gear screws.

Torque Value: 28 n•m [248 in-lb]

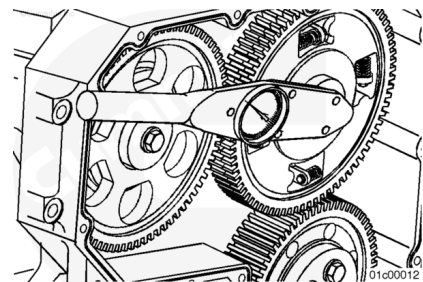
The valve camshaft gear will rotate slightly on the valve camshaft during this step.

Tighten the injector camshaft scissor gear retaining capscrew.

Torque Value:

1. 136 n•m [100 ft-lb]
2. Rotate an additional 90 degrees

Loosen the capscrew 360 degrees and tighten again using the torque-plus-angle method.

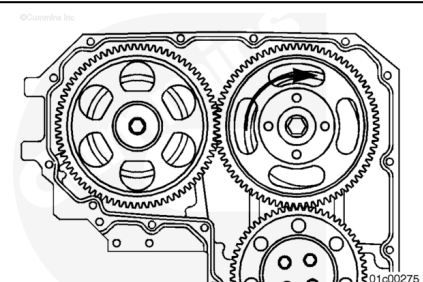


Torque Value:

1. 136 n•m [100 ft-lb]
2. Rotate an additional 90 degrees

Injector Camshaft Non-Scissor Gear

Use hand pressure **only** to rotate the injector camshaft gear **clockwise** until the gear train backlash is taken up. Obtain tooth-to-tooth contact between the crankshaft gear and the injector camshaft gear.



Tighten the injector camshaft non-scissor gear retaining capscrew.

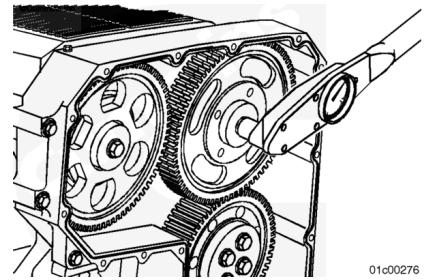
Torque Value:

1. 136 n•m [100 ft-lb]
2. Rotate 90 degrees.

Loosen the capscrew 360 degrees and tighten again using the torque-plus-angle method.

Torque Value:

1. 136 n•m [100 ft-lb]
2. Rotate 90 degrees.

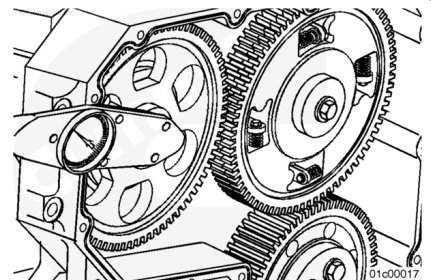


Valve Camshaft Scissor Gear

Tighten the valve camshaft scissor gear retaining capscrew.

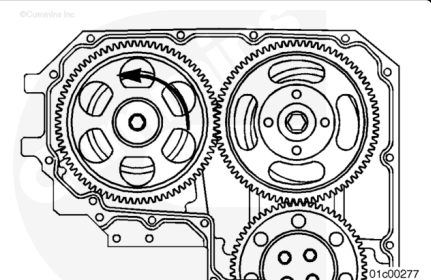
Torque Value: 102 n•m [75 ft-lb]

Rotate the valve camshaft gear mounting capscrew an additional 120 degrees.



Valve Camshaft Non-Scissor Gear

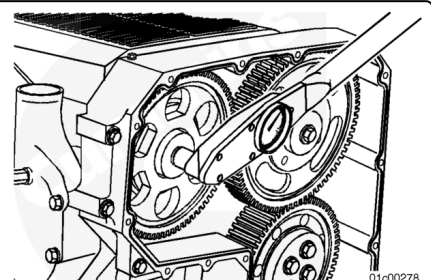
Use hand pressure **only** to rotate the valve camshaft gear **counterclockwise** until the backlash between the injector camshaft gear is taken up. Obtain tooth-to-tooth contact between the injector camshaft gear and the valve camshaft gear.



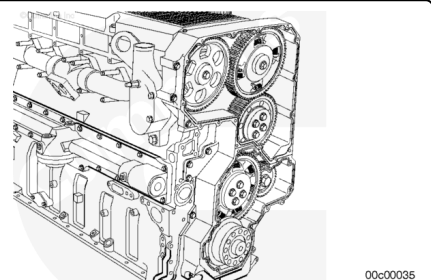
Tighten the valve camshaft non-scissor gear retaining capscrew.

Torque Value: 102 n•m [75 ft-lb]

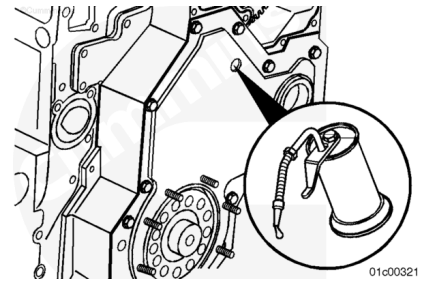
Rotate the valve camshaft gear mounting capscrew an additional 120 degrees.



Coat the entire gear train with clean 15W-40 oil.

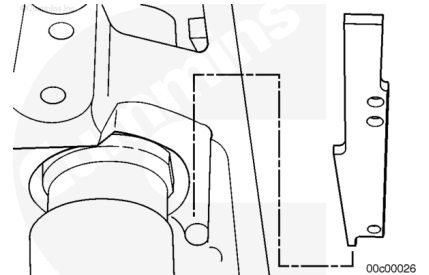


Use the access hole in the lower gear cover to lubricate the lower idler gear and crankshaft.

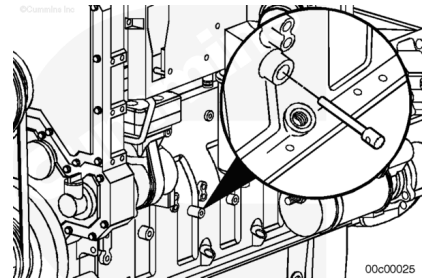


Remove the injector and valve camshaft timing wedges.

A sudden jerking motion upward is the most effective way to remove wedges.

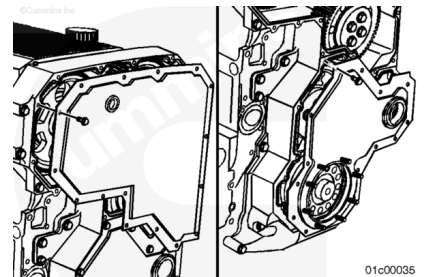


Remove the crankshaft locking pin and install the plug.

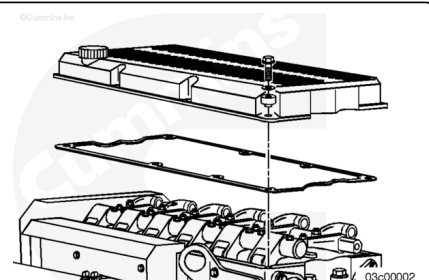


Finishing Steps

- Install the upper gear cover. Refer to Procedure 001-079 in Section 1. (</qs3/pubsys2/xml/en/procedures/10/10-001-079-tr.html>)
- Install the lower gear cover. Refer to Procedure 001-080 in Section 1. (</qs3/pubsys2/xml/en/procedures/10/10-001-080-tr.html>)



- Install the rocker lever cover and gasket. Refer to Procedure 003-011 in Section 3. (</qs3/pubsys2/xml/en/procedures/10/10-003-011-tr.html>)
- If coolant, oil, excessive fuel, or excessive black smoke has entered the exhaust system, the aftertreatment system **must** be inspected. Refer to Procedure 014-013 in



Section 14. (/qs3/pubsys2/xml/en/procedures/101/101-014-013-tr.html)

- Operate the engine and check for leaks.

⚠ WARNING ⚠

Batteries can emit explosive gases. To reduce the possibility of personal injury, always ventilate the compartment before servicing the batteries. To reduce the possibility of arcing, remove the negative (-) battery cable first and attach the negative (-) battery cable last.

- Connect the batteries. See equipment manufacturer service information.